A separation between a bright plasma core within the laser cone angle and a surrounding diffusely luminescent halo is vaguely perceptible also for ns breakdown at 1064 nm (Fig. 7 (d) and 355 nm (Fig. 7 (b), right column). However, the core is here not demarcated by scattered laser light because UV and IR wavelengths cannot be detected by the digital camera employed in our experiments. The brighter appearance of the core is rather due to an increase of plasma luminescence caused by a high volumetric energy density.

Generally, the extent of the luminescent halo beyond the plasma core is largest in forward direction. The existence of a halo outside the laser cone angle at large *NA* is indicative for a massive energy transport during breakdown. This transport is most likely mediated by energetic photons produced as bremsstrahlung from hot electrons in the string-like regions of primary energy deposition [238-240] as will be discussed in detail in section V.D.

Plasma features change for ns breakdown at moderate *NA*, as seen in Fig. 8 with photographs of plasmas produced by 6.8-ns, 355-nm pulses focused at *NA* = 0.3. Here the luminescent plasma region extends from the beam waist upstream towards the incoming laser beam, and it is largely confined within the laser cone angle. A bright spot demarcating the region of highest plasma energy density is located in the upstream part of each plasma, while the regions closer to the beam waist luminesce only weakly and

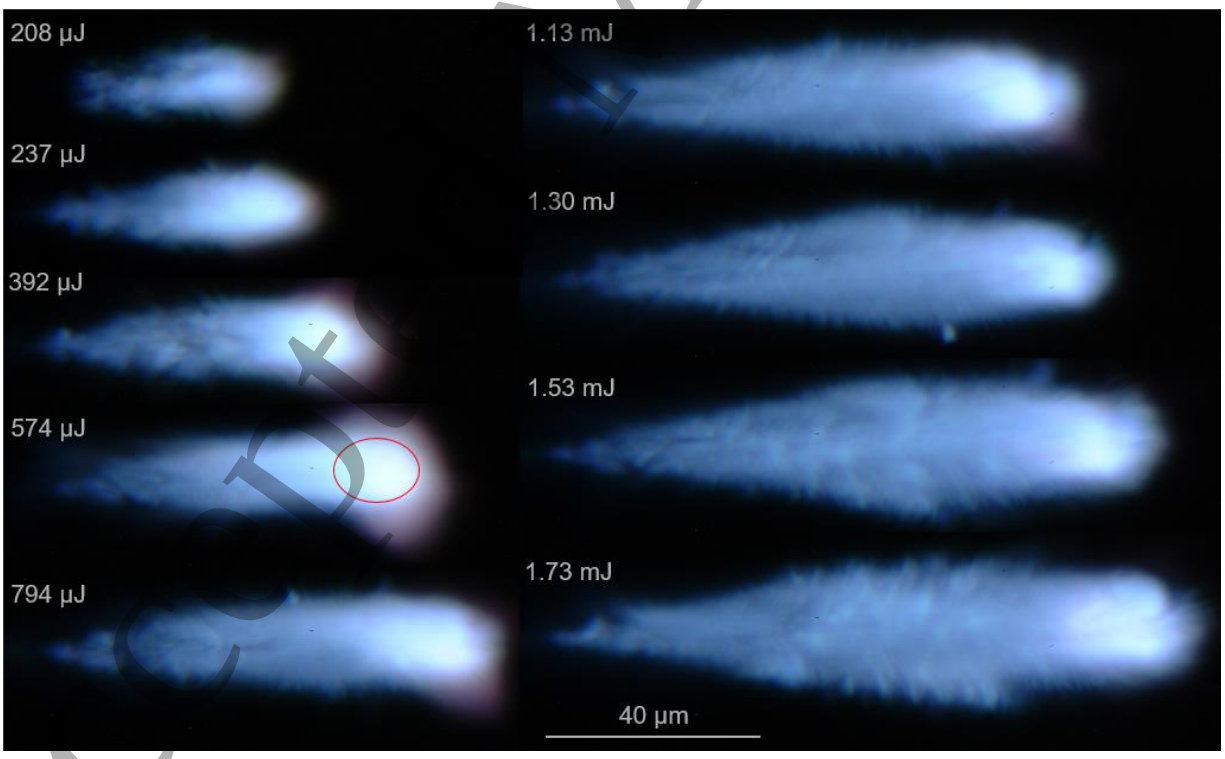

Figure 8 Side-view photographs of plasmas in water produced at *NA* = 0.3 by temporally smooth laser pulses of 6.8-ns duration, 355 nm wavelength and different pulse energies. The bubble threshold is E_{th} = 44 µJ, and the BPL threshold (50% breakdown probability) is 238 µJ. Laser light is incident from the right.

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exhibit a mottled substructure with fairly sharp borders. By contrast, the region around the hot spots is blurred and surrounded by a halo with colour transition from a whitish appearance in the hot spot to a reddish hue in the outer region. The hot spot and the halo are particularly pronounced for pulse energies between ≈ 250 µJ and ≈ 750 µJ. The reddish outer rim reaches partially into the region outside the laser cone angle and arises from ongoing luminescence during initial plasma expansion, when adiabatic cooling red-shifts the emission spectrum. For estimating the maximum energy density in the hot region within the cone angle, we assume that the volume marked by the red circle in the plasma produced with E_L = 574 µJ contains all energy deposited during the second half of the laser pulse. This way, we get that 287 µJ are deposited into a volume of 1100 µm^3, corresponding to an ellipsoid with half axes of 7.5 µm and 6 µm, which leads to a local energy density of 260 kJcm^{-3}.

4.4. Energy density, electron density, pressure, and temperature of luminescent plasmas

Figure 9 shows the procedure of determining the average plasma energy density in dependence of laser pulse energy on the examples of IR fs and UV ns pulses focused at *NA* = 0.8.

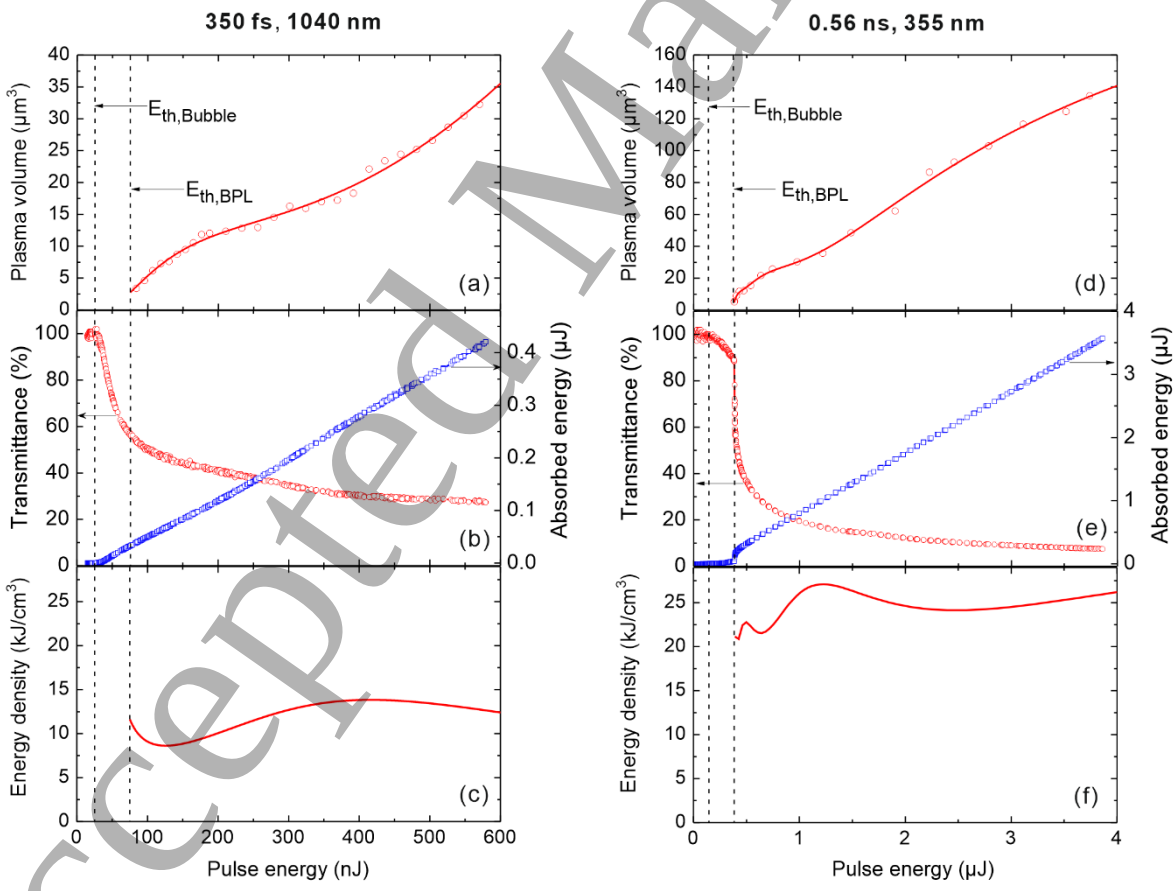

Figure 9 Determination of average plasma energy density $U_{avg} = E_{abs}/V_{plasma}$ for 1040-nm, 350-fs pulses (a-c), and 355-nm, 0.56-ns pulses (d-f) focused at *NA* = 0.8. The plasma volume V_{Plasma} (E_L) in (a) and (d) was determined from photographs, and the absorbed energy E_{abs}(E_L) in (c) and (e) was obtained from plasma transmittance T_{tra}(E_L)

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as $E_{abs} = E_L(1-T_{tra})$. The U_{avg} (E_L) curves in (c) and (f) were then obtained by dividing a fit curve through the $E_{abs}(E_L)$ data by the fit curve through the $V_{Plasma}(E_L)$ data.

The $U_{avg}(E_L)$ curves of Figs. 9 (c) and (f) refer to the high-density plasma regime because only luminescent plasmas could be evaluated. For the examples shown, U_{avg} varies little with pulse energy, and similar observations were also made for other wavelengths and pulse durations. Therefore, we determined mean values, $\bar{U}_{avg}$, to facilitate the comparison of plasma energy densities in different regions of the (τ_L, λ_L) space. The E_L range used for averaging starts for fs pulses at the onset of plasma luminescence ($3 \times E_{thB}$) and for ns pulses at $1.5 \times E_{thBPL}$. Its upper end is always given by the available laser pulse energy.

The $\bar{U}_{avg}$ values were used to derive plasma pressure and temperature with the help of the $p(U)$ and $T(U)$ diagrams of Fig. 3. This approach requires stress confined energy deposition, which applies for all pulse durations including ns breakdown because luminescent ns plasmas are much larger than the focal volume, as seen in Figs. 7 and 8. The movement of breakdown wave in upstream direction during the laser pulse further enhances stress confinement as it reduces the local energy deposition time below the laser pulse duration.

For fs plasmas, where only one set of free electrons is produced during breakdown, we used $\bar{U}_{avg}$ to determine the average electron density using Eq. (13) and compare it to the critical free-electron density ρ_{crit} at which the plasma frequency reaches the frequency of the laser light [Eq. (14)]. The resulting values for $\bar{U}_{avg}$, $\bar{P}_{avg}$, $\bar{T}_{avg}$, $\bar{\rho}_{c,avg}$ and ρ_{crit} are summarized in **Table 3**.

Plasma energy densities in luminescent plasmas are similar for fs and ns breakdown but $\bar{U}_{avg}$ varies with wavelength between ≈ 10 kJcm^{-3} and ≈ 40 kJcm^{-3} for both pulse durations. These values are 10-100 times larger than the energy density at the bubble threshold, which amounts to ≈ 1.2 kJ cm^{-3} for ns pulses [188] and ≈ 0.6 kJ cm^{-3} for fs breakdown [22]. Using the EOS date in Fig. 3, we see that the average plasma temperatures in the BPL regime range between 2800 and 10,100 K, and the corresponding pressures lie between 42 and 116 kbar. The $\rho_{c,avg}/\rho_{crit}$ ratio increases with wavelength from 0.37 at 347 nm through 2.94 at 520 nm to 4.82 at 1040 nm wavelength. Thus, for visible and IR wavelengths, the average free-electron density in luminescent fs plasmas produced at NA = 0.8 is supercritical.

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